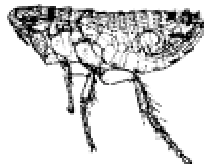
flea



NEWS 76 *June 2015*

http://esanetworks.org/

FLEA NEWS is a biannual newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as National Agricultural Library (NAL) Agricola, and National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Agricola or Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Contents

Editorial

Dear Flea News Reader,

In *Flea News* 71, I asked, why are there so many fleas? A more fundamental question is: how many flea species are there?

There are several layers to this apparently simple question: from simple tabulation (see the New Book Review in this *Flea News*), to the question of taxonomic validity, and finally to the underlying biology of fleas, both evolutionary and ecological.

The best estimates of flea diversity range around 2,000 flea species. No one knows the exact number, due to the currently dispersed flux of the taxonomic literature. That is why efforts to systematically centralize descriptions of diversity are critical to conserving the world's biological diversity.

Yours in fleas,
R.L. Bossard
Editor, Flea News

Announcements

Congratulations to Amoret Whitaker on her having been awarded the DipFMS (Diploma in Forensic Medical Sciences), awarded by The Society of Apothecaries of London, and the Doctor of Philosophy, for her dissertation in forensic entomology by School of Biomedical Sciences, King's College London.

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Congratulations to Terry Galloway upon his retirement from the Department of Entomology at the University of Manitoba. He is continuing his research on ectoparasites in his laboratory.

New Dissertations and Theses

Student: Abraham, Gifty.
Title: Target zero: Why states choose to eradicate infectious diseases and how they succeed
Institution: Dept. of Political Science, City University of New York
Degree: M.A.
Date: 2015
82 pages

Realism has remained the dominant paradigm within international relations for most of the modern era, emphasizing the competitive nature of the international arena and the unlikeliness of states to within it to cooperate. The attempts and further still, successes, by states to eradicate infectious diseases--which remain among the most cooperative enterprises--present a number of challenges to realism's assumptions, particularly with respect to the unlikely world historical-times during which the eradication campaigns took place. As such, a two-part puzzle arises. First, why would states, which are natural competitors, cooperate to eradicate infectious diseases given structural and situational incentives not to do so? Second, when states choose to eradicate infectious diseases, what accounts for success? I tackle these questions in turn, proposing three factors: reputational benefits, rational self-interest, and the interplay of state and non-state actors, all of which, in addition to international institutions help explain the involvement and cooperation of state actors in eradication efforts, particularly at those critical periods. To answer the second question, I offer a five-point rubric which features necessary, if insufficient conditions for the success of any eradication campaign. They are: ways to cure, treat or manage the disease and its spread; monitoring programs; political will; community engagement; and agencies tasked, and adequately equipped to deal with the disease. I answer the proposed questions and apply both sets of factors using three case studies; they include the successful case of smallpox eradication, the first failed campaign against malaria of the 1950s and the modern ongoing effort, and the campaign against guinea worm disease today.

Student: Ellis, Vincenzo A.
Title: Relationships of Haemosporidian parasites to populations of their avian hosts in eastern North America
Institution: Biology, University of Missouri - Saint Louis
Degree: Ph.D.
Date: 2015
108 pages

Avian Haemosporida are common, vector-transmitted blood parasites of birds throughout the world. During my dissertation research, I explored how multiple host species respond immunologically to natural infections in the wild (Chapter 1) and to experimental infections in the laboratory (Chapter 2). Despite their tractability as a model host-parasite system and a burgeoning literature on avian Haemosporida, little is known about how their populations interact across large areas (hereafter "regions"). I present data from parasite surveys of birds across eastern North America suggesting that continental parasite populations track host populations across the region, but also that the host breadth of a parasite can be variable across space and time (Chapter 3). Parasite lineages replace each other spatially within a host population, likely due to interspecific parasite competition mediated by host immune systems (Chapter 3). Parasite prevalence is positively related to host abundance within local assemblages (Chapter 4), but within host species across their ranges, prevalence does not vary with abundance (Chapters 3 and 4). Finally, a 12 year survey of parasites and their hosts in the Missouri Ozarks demonstrates that parasite populations vary through time, and that this variability is related to host breadth--specialist parasites (i.e., parasites infecting primarily one host) were more variable than generalist parasites (i.e., parasites infecting multiple hosts; Chapter 5). Overall my dissertation work contributes to the natural history and ecology of avian Haemosporidian parasites and their avian hosts, and to host-parasite ecological and evolutionary theory.

Student: Grossmann, Nárjara Veras
Title: Relações Parasito-Hospedeiro de Endo e Ectoparasitas em Pequenos Mamíferos no Cerrado do Brasil Central [Parasite-Host Relationships of Endo and Ectoparasites in Small Mammals in the Cerrado of Central Brasil]
Institution: Universidade de Brasília, Instituto de Ciências Biológicas
Degree: Doctoral
Date: 2015

A importância dos parasitas nos sistemas ecológicos vem sendo estudada nos últimos anos. No entanto, classificar a diversidade existente ainda é necessário para que estas pesquisas avancem. O presente trabalho teve como objetivo identificar os parasitos que acometem roedores de uma área aberta do Cerrado, além dos seus padrões epidemiológicos e ecológicos antes e após um incêndio. Ao longo de um ano, roedores foram capturados em dois grids e duas linhas de coleta na Estação Ecológica de Águas Emendadas. Os dados obtidos nos grids foram utilizados nas análises populacionais dos hospedeiros e os dados das linhas acessórias foram utilizados para identificar e mensurar os parasitos. Dentre os endoparasitas as seguintes espécies foram encontradas: *Stilestrongylus freitasi*, *S. stilesi, Hassalstrongylus* sp., *Syphacia alata*, *Syphacia criceti*, *S. evaginata*, *S. obvelata*, *S. venteli*, *Protospirura n. criceticola*, *Hymenolepis* sp. e *Taenia* sp., *Pterygodermatites (P.) zygodontomys* e uma espécie de nova de *Pterygodermatites*. Siphonaptera (*Polygenis* sp.), Phthiraptera (*Hoplopleura* sp.) e Acari (*Ixodes* sp., e as famílias Dermanyssidae; Laelapidae; Macrochelidae e Macronyssidae) compõem as amostras de ectoparasitas. Para algumas espécies esses são os primeiros registros para o DF e para os hospedeiros. Padrões de distribuição de ectoparasitas revelaram uma distinção na ocupação do corpo do hospedeiro entre ácaros e piolhos. Dentre os hemoparasitos foram identificados *Trypanosoma*, *Mycoplasma* hemotrófico e uma espécie não identificada. De forma geral *Necromys lasiurus* apresentou maior prevalência, riqueza e diversidade de parasitos comparados com *Calomys tener* e *C. expulsus*. Após o incêndio os parasitos de *Calomys* foram mais freqüentes e mais diversos comparados com o período antes do fogo, ao passo que em *Necromys*, houve perda tanto em diversidade quanto prevalência. As variações observadas nos padrões dediversidade e prevalência parasitária não podem ser diretamente atribuídas ao fogo, portanto a coleta adicional de dados poderá esclarecer este fato.

[The importance of parasites in ecological systems has been studied in recent years. However, classifying the diversity is still needed for this research to advance. This study aimed to identify the parasites that affect rodents in an open area of the Cerrado, in addition to their epidemiological and ecological patterns before and after a fire. Over a year, rodents were captured in two grids and two collection rows in Emendadas Water Ecological-Station. Data from grids were used in population analyzes of hosts and data of accessory lines were used to identify and measure the parasites. Among the endoparasites the following species were found: *Stilestrongylus freitasi*, *S. stilesi*, *Hassalstrongylus* sp., *Syphacia alata*, *Syphacia criceti*, *S. evaginata*, *S. obvelata*, *S. venteli*, *Protospirura n. criceticola*, *Hymenolepis* spp. and *Taenia* sp., *Pterygodermatites (P.) Zygodontomys* and a kind of new *Pterygodermatites*. Siphonaptera (*Polygenis* sp.), Phthiraptera (*Hoplopleura* sp.) and Acari (*Ixodes* sp., and Dermanyssidae families; Laelapidae; Macrochelidae and Macronyssidae) make up samples of ectoparasites. For some species these are the first records for the DF and the hosts. Ectoparasites distribution patterns revealed a distinction in the host body occupancy between mites and lice. Among the blood parasites *Trypanosoma* were identified, *Mycoplasma* hemotrófico and one unidentified species. Overall *Necromys lasiurus* showed higher prevalence, richness and diversity of parasites compared with *Calomys tener* and *C. expulsus*. After the fire the parasites *Calomys* were more frequent and more diverse compared with the period before the fire, whereas in *Necromys*, there was a loss both in diversity and prevalence. The variations observed in diversity standards and parasite prevalence can not be directly attributed to the fire, so the additional data collection may account for this fact. [Editor's abstract]

Student: Maschek, Kevin Steven.
Title: The efficacy of a spot-on pesticide against ectoparasites affecting poultry in Mississippi
Institution: Veterinary Medicine, Mississippi State University
Degree: M.S.
Date: 2015
44 pages

Ectoparasites are a common occurrence in backyard and commercial chickens and present a severe health challenge in certain circumstances. The accurate identification of the parasite is important in properly treating and preventing re-infestation. The feeding behavior of the pest species and management factors of the chicken affect outcomes of different types of treatment strategies. Most commercial products come in dusts, fogs, mists and sprays. The novel use of a topical spot-on pesticide can provide efficient adulticidal activity against stick tight fleas (*Echidnophaga gallinacea*) and bed bugs (*Cimex lectularius*) associated with poultry in Mississippi.

Student: Sadaria, Akash Mahendra
Title: Fate of Six Neonicotinoids During Full-scale Wastewater Treatment and Passage Through an Engineered Wetland
Institution: Civil and Environmental Engineering, Arizona State University
Degree: M.S.
Date: 2015
42 pages

Six high-production-volume neonicotinoids were traced through a municipal wastewater treatment plant (WWTP) and engineered wetland located downstream, in a study motivated by reports on these insecticides posing threats to non-target invertebrate species and potentially playing a role in the global honeybee colony collapse disorder. An array of automated samplers was deployed in a five-day monitoring campaign and resultant flow-weighted samples were analyzed by liquid chromatography tandem mass spectrometry (LC-MS/MS) using the isotope dilution method. Concentrations in WWTP influent and effluent were 54.7 ± 2.9 and 48.6 ± 2.7 ng/L for imidacloprid, respectively, and 3.7 ± 0.3 and 1.8 ± 0.1 ng/L for acetamiprid, respectively. A mass balance over the WWTP showed no ($p = 0.09$, CI = 95%) removal of imidacloprid, and 56 ± 6% aqueous removal of acetamiprid. In the constructed wetland downstream, a lack of removal was noted for both imidacloprid (from 54.4 ± 3.4 ng/L to 49.9 ± 14.6 ng/L) and acetamiprid (from 2.00 ± 0.03 ng/L to 2.30 ± 0.21 ng/L). Clothianidin was detected only inconsistently in the WWTP and wetland (>2 to 288 ng/L; 60% detection frequency), whereas thiamethoxam (<10 ng/L), thiacloprid (<2 ng/L), and dinotefuran (<180 ng/L) were not detected at all. Thus, imidacloprid and acetamiprid were identified as recalcitrant sewage constituents (estimated U.S. WWTP discharge of 1920- 4780 kg/y) that persist during conventional wastewater treatment to enter U.S. surface waters at potentially harmful concentrations.

Student: White, Alison.
Title: Linking human-disturbed landscapes with pathogen prevalence in wildlife: A meta-analysis

Institution: Biology, Colorado State University
Degree: M.S.
Date: 2015
60 pages

The percentage of earth disturbed by humans is rapidly growing. In a review of the available literature in wildlife disease ecology we performed a meta-analysis to determine if human disturbed landscapes increased prevalence of pathogens and parasites, compared to undisturbed landscapes. We analyzed a total of 68 cases of host-pathogen and host-parasites reported in 34 publications. We carried out analyses at two levels: 1.) studies reporting prevalence values for both disturbed and undisturbed landscapes, of which 46 cases within 13 published studies were included in the final analysis and 2.) studies only reporting differences in disturbed and undisturbed landscapes (increase, decrease, varied, or no change) without published prevalence data (68 cases within 36 studies). Overall, we found that disturbed landscapes had higher pathogen prevalence. We reviewed potential indirect drivers (types of landscapes), direct drivers (features associated with landscapes), and mechanisms (changes in ecology caused by the indirect and direct drivers) that may account for the increase in pathogen prevalence between the landscapes. High pathogen prevalence in wildlife living in disturbed landscapes may serve as an indicator of the negative consequences of unsustainable human development. Having this understanding will enable wildlife managers to produce sustainable development solutions that will improve their predictions of infection and reduce prevalence of harmful pathogens in sensitive populations.

Student: Zhao, Jianguo
Title: Biologically inspired approach for robot design and control
Institution: Electrical Engineering, Michigan State University
Degree: Ph.D.
Date: 2015
174 pages

Robots will transform our daily lives in the near future by moving from controlled industrial lines to unstructured and uncertain environments such as home, offices, or outdoors with various applications from healthcare, service, to defense. Nevertheless, two fundamental problems remain unsolved for robots to work in such environments. On one hand, how to design robots, especially meso-scale ones with sizes of a few centimeters, with multiple locomotion abilities to travel in the unstructured environment is still a daunting task. On the other hand, how to control such robots to dynamically interact with the uncertain environment for agile and robust locomotion also requires tremendous efforts. This dissertation tries to tackle these two problems in the framework of biologically inspired robotics. On the design aspect, it will be shown how biologically principles found in nature can be used to build efficient meso-scale robots with various locomotion abilities such as jumping, wheeling, and aerial maneuvering. Specifically, a robot (MSU Jumper) with continuous jumping ability will be presented. The robot can achieve the following three performances simultaneously. First, it can perform continuous steerable jumping based on the self-righting and the steering capabilities. Second, the robot only requires a single actuator to perform all the functions. Third, the robot has a light weight (23.5 g) to reduce the damage from landing impacts. Based on the MSU Jumper, a robot (MSU Tailbot) with multiple locomotion abilities is discussed. This robot can not only wheel on the ground but also jump up to overcome obstacles. Once leaping into the air, it can also control its body angle using an active tail to dynamically maneuver in mid-air for safe landings. On the control aspect, a novel non-vector space control method that formulates the problem in the space of sets is presented. This method can be easily applied to vision based control by considering images as sets. The advantage of such a method is that there is no need to extract and track features during the control process, which is required by traditional methods. Based on the non-vector space approach, the compressive feedback is proposed to increase the feedback rate and reduce the computation time. This method is ideal for the control of meso-scale robots with limited sensing and computation ability. The bio-inspired design illustrated by the MSU Jumper and MSU Tailbot in this dissertation can be applied to other robot designs. Meanwhile, the non-vector space control with compressive feedbacks lays the foundation for the control of high dynamic meso-scale robots. Together, the biologically inspired method for the design and control of meso-scale robots will pave the way for next generation bio-inspired, low cost, and agile robots.

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Meetings

The 46th Annual Society for Vector Ecology Conference will be held September 27 through October 1, 2015, at the Embassy Suites in Albuquerque, New Mexico. Poster submissions are solicited with a deadline of August 15th . Further information and forms are available at the SOVE website, www.sove.org.

New Book Review

Title: World Species Index: Siphonaptera: A Comprehensive Catalog of Fleas of the World
Author: World Species Index
Publisher: CreateSpace Independent Publishing Platform
Publication: July 4, 2013, ISBN-10: 1490901558, ISBN-13: 978-1490901558
62 pages, paperback

The *World Species Index: Siphonaptera: A Comprehensive Catalog of Fleas of the World* is a listing of flea species in the world, with their taxonomic authority and year. There are similar books for other taxa (for example, the Odonata) that the public seems to find useful. Such a listing, if accurate, should be eminently useful for anyone working with fleas. However, closer examination provokes a certain skepticism. There is no author of this book, at least, not in the conventional sense. The book seems to have been compiled anonymously by a group of writers, not scientists.

The general controversy about the appropriation of anonymously written information from open-access sources such as "wikipedias" available in electronic form from the internet, and its publication in paper books is beyond the scope of this review. Suffice it to say, a variety of publishers are finding a ready audience for print editions of well-written and informative material which might otherwise vanish into the electronic ether, but on the other hand, such community-created material often is ridden with errors, and lacking in sufficient attribution, editing, or critical review, with no one actually taking responsibility for the contents.

Ironically, this listing of around 2000 flea species, and their authorities, has no authority. This brings up a pointed question. Why are flea experts not writing books like this? According to my understanding, Dr. Robert Lewis, previous editor of *Flea News*, compiled a numbered list of flea species for many years. However, this list has never been published. Similarly, there are websites that keep extensive lists of flea species, genera, and families with taxonomic authorities, but again these lists are unpublished.

Though each issue of *Flea News* provides abundant evidence every year of enormous amounts of publications on fleas, the research is highly fragmented, often redundant. There is little synthesis going on. This book is one that anyone interested in fleas would find valuable. We need more synthetic reviews of fleas' ecology, evolution, life history, distribution and abundance, and disease and pathogens. A list of all the flea species is a start.

R.L. Bossard

Featured Research

Linardi P., Beaucournu J.-C., D'Avelar & Belaz S., Notes sur le genre *Tunga* Jarocki, 1838 (Siphonaptera- Tungidae). II. Neosomes, morphology, classification and others taxonomic notes. Parasite, 2014, X : x-x. in press.

Beaucournu J.-C., Moreno L. & González-Acuña D. Fleas (Insecta - Siphonaptera) of Chile: a revision. Zootaxa, 2014, 3900 (2): 151 – 203.

Beaucournu J.-C. - Le Professeur Jean-Marie Doby (1925-1999), le Parasitobus, et les autres...ou Petite chronique de la Parasitologie à la Faculté de Médecine et de Pharmacie de Rennes de 1956 à 2007. Conférences rennaises d'Histoire de la Médecine et de la Santé, 2014, Vol. 11.

Quetglas J., Nogueras J., Ibañez C. & Beaucournu J.-C.. Presencia en la Peninsula Iberica de una pulga africana de murcielagos : *Rhinolophopsylla unipectinata arabs* (Siphonaptera: Ischnopsyllidae) y otras nuevas citas de pulgas de Murcielagos para Españé y Marruecos. Galemys, 2015, in press.

Sanchez J., Beaucournu, J.-C., Lareschi, M. Revision of the fleas of the genus *Plocopsylla* belonging to the "*angusticeps-lewisi*" complex in the Andean biogeographic region, with the description of a new species. Medical and Veterinary Entomology, 2015, 29 , 147–158. doi: 10.1111/mve.121

In Argentina, the Andean biogeographic region accommodates the most diverse population of fleas in the country. The Craneopsyllinae (Siphonaptera: Stephanocircidae) represent one of the most commonly found subfamilies in this region and show some endemism and high diversity. *Plocopsylla* is the most diverse genus of Craneopsyllinae; it includes 10 species mainly distributed in the Patagonian subregion, which parasitize sigmodontine rodents (Rodentia: Cricetidae). We describe and illustrate the morphology of the aedeagus in species of *Plocopsylla* that belong to the '*angusticeps–lewisi*' complex. This character is of diagnostic value in differentiating among species. A new species of this complex, *Plocopsylla (Plocopsylla) linardii* sp. n., is described and identified by the shape and chaetotaxy of the distal arm of sternite

IX, as well as by the shape of the median dorsal lobe of the aedeagus. New host associations for this complex and range extensions for most of its species are reported. *Plocopsylla (P.) silewi* is recorded for the first time in Argentina. The southern limits of the distributions of *Plocopsylla (P.) lewisi* and *Plocopsylla (P.) wilesi* are extended to Santa Cruz Province. The *angusticeps–lewisi* complex is found for the first time in San Juan Province. The information may be useful in epidemiological studies of flea-borne diseases.
Key words. Craneopsyllinae, Cricetidae, Sigmodontinae, Siphonaptera, Stephanocircidae, aedeagus, Argentina, Patagonia.

Belaz, S., E. Gay, F. Robert-Gangneux, J. -C Beaucournu, and C. Guiguen. 2015. Tungiasis Outbreak in Travelers from Madagascar. Journal of Travel Medicine. Volume 22, Issue 4, July/August 2015, Pages: 263–2662 JUN 2015, DOI: 10.1111/jtm.12217

López-Berrizbeitia, M. F., J. Sanchez, M. M. Díaz, R. M. Barquez, and M. Lareschi. 2015. Redescription of *Neotyphloceras crassispina hemisus* Jordan (Siphonaptera: Ctenophthalmidae: Neotyphloceratini). Journal of Parasitology 101 (2): 145-149.

Neotyphloceras crassispina hemisus Jordan is redescribed from male and female specimens collected at the type locality (Otro Cerro, Catamarca Province, Argentina) and nearby localities. New diagnostic morphological characters for both sexes are provided, which include the shape of the upper lobe of the fixed process of clasper, the crochet of the aedeagus and the shape and chaetotaxy of the distal arm of sternum IX for males, and the contour of the distal margin of sternum VII for females.

Cameron, S. L. 2015. The Complete Mitochondrial Genome of a Flea, *Jellisonia amadoi* (Siphonaptera: Ceratophyllidae). Mitochondrial DNA 26 (2): 289-290.

Dryden, MW. 2014. Spotlight on Research: How residual speed of kill affects flea control in dogs and cats. Vet Med 109(7):1-4.

Dryden MW, Smith V, Chwala M, Jones E, Crevoiserat L, McGrady JC, Foley KM, Paon PR, Hawkins A, Carithers D. Evaluation of afoxolaner chewables to control flea populations in naturally infested dogs in private residences in Tampa FL, USA. Parasites & Vectors 8(1):286, 2015.

Background: A study was conducted to evaluate the effectiveness of afoxolaner chewables to control flea populations in naturally infested dogs in private residences in Tampa FL, USA. Evaluations of on-animal and premises flea burdens, flea sex structure and fed-unfed premises flea populations were conducted to more accurately assess flea population dynamics in households.
Methods: Thirty seven naturally flea infested dogs in 23 homes in Tampa, FL were enrolled in the study and treated with afoxolaner chewables. Chewables (NexGard® Chewables; Merial) were administered according to label directions by study investigators on study day 0 and once again between study days 28 and 30. Flea infestations on pets were assessed using visual area thumb counts and premises flea infestations were assessed using intermittent-light flea traps on days 0, 7, 14, 21, and once between study days 28–30, 40–45, and 54–60.
Results: Within 7 days of administration of afoxolaner chewable tablets, flea counts on dogs were reduced by 99.3 %. By one month post-treatment, total flea counts on dogs were reduced by 99.9 %, with 97.3 % (36/37) of the dogs being flea free. Following the second dosing on study day 28–30, total on-dog flea burden was reduced by 100 % on days 40–45 and 54–60. On day 0, the traps collected a geometric mean of 18.2 fleas. Subsequent reductions in emerging flea populations were 97.7 and 100

% by days 28–30 and 54–60, respectively. There were 515 total fleas (*Ctenocephalides felis felis*) collected in the intermittent light flea traps on day 0, and 40.4 % of those fleas displayed visual evidence of having fed. Seven days after initial treatment, only 13.1 % of the fleas contained blood and by day 14 only 4.9 % of the fleas collected in traps displayed evidence of having fed. On day 0, prior to treatment, 60 % of the unfed fleas collected in intermittent-light flea traps were females, but by days 28–30, unfed males accounted for 78 % of the population.
Conclusions: This in-home investigation conducted during the summer of 2014 in subtropical Tampa, FL demonstrated that afoxolaner chewables rapidly and effectively eliminated flea populations in infested dogs and homes.

Dryden, MW, Smith V, Hodgkins E. Varloud M. Residual Adulticidal Activity of a Dinotefuran-pyriproxyfen Topical Spot-on Formulation Applied to Dogs against Weekly Infestations with the KS1 Flea Strain. Intern. J. Appl. Res. Vet. Med. "in-press"

Dryden, M. W., Smith, V., Bennett, T., Math, L., Kallman, J., Heaney, K., & Sun, F. (2015). Efficacy of fluralaner flavored chews (Bravecto®) administered to dogs against the adult cat flea, *Ctenocephalides felis felis* and egg production. *Parasites & vectors 8*(1), 1-7.

David A. Eads, Dean E. Biggins, Michael F. Antolin, Dustin H. Long, Kathryn P. Huyvaert, and Kenneth L. Gage. 2015. Prevalence of the Generalist Flea *Pulex simulans* on Black-tailed Prairie Dogs (*Cynomys ludovicianus*) in New Mexico, USA: The Importance of Considering Imperfect Detection. Journal of Wildlife Diseases 51(2):498-502.

If a parasite is not detected during a survey, one of two explanations is possible: the parasite was truly absent or it was present but not detected. We fit occupancy models to account for imperfect detection when combing fleas (Siphonaptera) from black-tailed prairie dogs (*Cynomys ludovicianus*) during June–August 2012 in the Vermejo Park Ranch, New Mexico, USA. With the use of detection histories from combing events during monthly trapping sessions, we fit occupancy models for two flea species: *Oropsylla hirusta* [spelled *Oropsylla hirsuta* – Editor's note] (a prairie dog specialist) and *Pulex simulans* (a generalist). Detection probability was <100% for both species and about 21% lower for *P. simulans*. *Pulex simulans* may be especially difficult to detect because it is about half the size of *O. hirusta* [spelled *O. hirsuta* – Editor's note] . Monthly occupancy (prevalence) for *P. simulans* was estimated at 24% (June, 95% confidence interval=19–30), 39% (July, 32–47), and 56% (August, 49–64) in new prairie dog colonies, and 43% (32–54), 61% (49–71), and 79% (70–87) in old colonies. These results suggest *P. simulans* can attain high prevalence on prairie dogs, especially in old colonies. If *P. simulans* is highly prevalent on prairie dogs, it may serve as a "bridge vector" between *Cynomys* and other mammalian hosts of the plague bacterium *Yersinia pestis*, and even function as a reservoir of *Y. pestis* between outbreaks.

Eads, D. A. and Biggins, D. E. (2015), Plague bacterium as a transformer species in prairie dogs and the grasslands of western North America. Conservation Biology. doi: 10.1111/cobi.12498

Invasive transformer species change the character, condition, form, or nature of ecosystems and deserve considerable attention from conservation scientists. We applied the transformer species concept to the plague bacterium *Yersinia pestis* in western North America, where the pathogen was introduced around 1900. *Y. pestis* transforms grassland ecosystems by severely depleting the abundance of prairie dogs (*Cynomys* spp.) and thereby causing declines in native species abundance and diversity, including threatened and endangered species; altering food web connections; altering the import and export of nutrients; causing a loss of ecosystem resilience to encroaching invasive plants; and modifying prairie dog burrows. *Y. pestis* poses an important challenge to conservation biologists because it causes trophic-level perturbations that affect the stability of ecosystems. Unfortunately, understanding of the effects of *Y. pestis* on ecosystems is rudimentary, highlighting an acute need for continued research.

Ezquiaga, M. C., P. M. Linardi, D. M. De Avelar, and M. Lareschi. 2015. A New Species of *Tunga* Perforating the Osteoderms of its Armadillo Host in Argentina and Redescription of the Male of *Tunga terasma*. Medical and Veterinary Entomology 29 (2): 196-204.

A new species of *Tunga* (Siphonaptera: Tungidae) collected from armadillos in Argentina is described. The new species is characterized by large and pigmented eyes, the presence of two bristles on antennal segment II, two bristles at the base of the maxilla, and a discoid neosome compressed anteroposteriorly. The gravid female is located in the carapace of the host, perforating the osteoderms. The new species resembles *Tunga penetrans* and *Tunga terasma* in general appearance. However, it differs by the greater anteroposterior compression of the neosome, a more angular head, and a manubrium with a pointed proximal end and convex ventral margin (the proximal end of the manubrium is rounded or slightly pointed in *T. terasma*, and the ventral margin is straight in both *T. penetrans* and *T. terasma*). In addition, specimens of *T. penetrans* have more bristles in antennal segments II and III, and lack bristles in the posterior tibia. This is the first report of a species of *Tunga* perforating the osteoderms of its host and thereby showing a high degree of specialization. *Tunga terasma* is recorded for the first time in Argentina; the male is described again and the characteristics of the species amended. This information may be useful in epidemiological studies of diseases caused by species of *Tunga*.

Key words. *Chaetophractus vellerosus*, *Zaedyus pichiy*, Dasypodidae, Siphonaptera.

Huang, D. 2015. *Tarwinia australis* (Siphonaptera: Tarwiniidae) from the Lower Cretaceous Koonwarra Fossil Bed: Morphological Revision and Analysis of its Evolutionary Relationship. Cretaceous Research 52 (PB): 507-515.

Tarwinia australis, from the Lower Cretaceous Koonwarra fossil bed, Victoria, South Australia, is the first described Mesozoic flea. It was suggested to have a unique morphology that differs from all other known Mesozoic giant fleas by having a laterally-flattened body and peculiar tibial ctenidia. It represents an extinct family, Tarwiniidae, among the three major Mesozoic monster flea groups. Re-examination of the holotype reveals that this Southern Hemisphere ferocious bloodsucker bears different morphological details from those described previously. *Tarwinia australis* definitely bears elongate siphonate mouth-parts and a relatively compact antenna with 15 flagellomeres. Its legs are slender, elongate, and armed with pseudocombs-like ctenidia on all tibiae. The abdomen is covered with posteriorly-directed setal rows, and with a posteriorly-located pygidium and exposed male genitalia. It differs from pseudopulicids mainly on the basis of characters of tibial ctenidia that probably indicate a very different host association.

Kowalski, Krzysztof; Bogdziewicz, Micha; Eichert, Urszula; Rychlik, Leszek. 2015. Sex differences in flea infections among rodent hosts: is there a male bias? Parasitology Research Volume: 114 Issue: 1

Pages: 337-341. DOI: 10.1007/s00436-014-4231-z

Recognizing patterns of parasite distribution among wildlife hosts is of major importance due to growing risk of transmission of zoonotic diseases to humans. Thus, sex-dependent parasite distribution in higher vertebrates is extensively studied, and males are often found more parasitized than females. Male-biased parasitism may be the result of weaker immunocompetence of male hosts owing to the immunosuppressive effect of androgens. Moreover, larger hosts (males) may demonstrate higher parasite infestation levels than smaller individuals (females), as they constitute a better nutritional resource for parasites and provide them with a greater variety of niches. In the present work, we investigated sex-dependent patterns of flea distribution among three common rodent species (*Apodemus agrarius*, *Apodemus flavicollis*, and *Myodes glareolus*). We hypothesized that males have a higher flea infestation than females. We confirm male-biased parasitism in *A. agrarius* and *M. glareolus*, but not in *A. flavicollis*. Additionally, flea infestation increased with body mass in *A. agrarius*, but not in *A. flavicollis* and *M. glareolus*. The detected differences in parasite distribution among sexes are probably the result of immunosuppressive effects of androgens and spatial behavior of males.

Charlotte Lechat, Noémie Siméon, Olivier Pennant, Loïc Desquilbet, Sabine Chahory, Christophe Le Sueur and Jacques Guillot. 2015. Comparative evaluation of the prophylactic activity of a slow-release insecticide collar and a moxidectin spot-on formulation against *Thelazia callipaeda* infection in naturally exposed dogs in France. Parasites & Vectors (2015) 8:93.

Background: The relative efficacy of a collar containing 10% imidacloprid and 4.5% flumethrin (Seresto®, Bayer HealthCare Animal Health) and a spot on formulation containing 10% imidacloprid and 2.5% moxidectin (Advocate®, Bayer HealthCare Animal Health) was evaluated as a control measure to prevent canine thelaziosis in dogs in an endemic area of France.
Findings: Ninety-six privately-owned dogs were enrolled in the multicentre, controlled study. Before summer (the period of transmission by fruit flies), dogs were allocated to one of three groups: Group A (n = 36)- treated once with a collar containing 10% imidacloprid and 4.5% flumethrin; Group B (n = 33)- treated every month for 8 months with a spot-on containing imidacloprid 10% and moxidectin 2.5%; and Group C (n = 27)- untreated control animals. Dogs were regularly subjected to ocular examination in order to assess *Thelazia callipaeda* infection. During the trial, *T. callipaeda* nematodes were detected in 12 (33%) collared dogs (group A) whereas no eyeworm could be found in dogs who received a monthly spot on application of moxidectin (group B). In the control group, 8 (30%) dogs became infected.
Conclusions: The monthly application of a spot on formulation containing 10% imidacloprid and 2.5% moxidectin was shown to be highly effective in preventing *T. callipaeda* infection in a population of dogs living in an endemic area in France. On the contrary, the slow-release collar tested in this study did not display any protection against canine thelaziosis.
Keywords: *Thelazia callipaeda*, Eyeworm, Prevention, Dog, France

Mescht, L., Matthee, S., & Matthee, C. A. (2015). A genetic perspective on the taxonomy and evolution of the medically important flea, *Dinopsyllus ellobius* (Siphonaptera: Dinopsyllinae), and the resurrection of *Dinopsyllus abaris*. Biological Journal of the Linnean Society, first published online: 8 JUL 2015 DOI: 10.1111/bij.12615

Dinopsyllus ellobius is considered a common and widespread flea in southern Africa and can act as a vector for plague. Due to differences in the interpretation of geographical variation in male sternite VIII, the taxonomy of the species is characterized by uncertainty. In an attempt to provide a better understanding of the systematics of *D. ellobius*, and also to provide new insights into the mechanisms that play a role in the diversification of the taxon, we sampled 830 small mammals at 31 localities throughout the distribution range of the parasite. Mitochondrial cytochrome *c* oxidase subunit II (COII) sequence data were generated for 151 *D. ellobius* specimens from 19 positive localities and this data set was supplemented with partial data derived from an elongation factor 1 alpha (EF1-α) intron. A parsimony haplotype network of the mtDNA and a Splitstree analyses revealed the existence of two genetic clades that were separated by 5.53% (± 1.78) sequence divergence. Although the nuclear DNA data were unresolved, significant size differences were detected between head, coxa, femur and tibia lengths of male individuals belonging to the two mtDNA lineages. The exact mechanisms that could have caused the diversification among lineages are not clear but the two lineages seem to be geographically separated and may have different ecological requirements. The present study strongly supports the notion that the two lineages are representative of *D. ellobius* (probably more associated with the host body and better adapted to mesic conditions) and *Dinopsyllus abaris* (probably more associated with the host nest and diverse climatic conditions) as originally proposed based on the single morphological character confined to male sternite VIII.

van der Mescht, L., S. Matthee, and C. A. Matthee. 2015. Comparative Phylogeography between Two Generalist Flea Species Reveal a Complex Interaction between Parasite Life History and Host Vicariance: Parasite-Host Association Matters. BMC evolutionary biology, 15(1), 105.

GUSTAVO F. PAZ, DANIEL M. AVELAR, ILKA A. REIS, AND PEDRO M. LINARDI. 2015. Dynamics of *Ctenocephalides felis felis* (Siphonaptera: Pulicidae) Infestations on Urban Dogs in Southeastern Brazil. J. Med. Entomol. 1–6 (2015); DOI: 10.1093/jme/tjv071

The cat flea, *Ctenocephalides felis felis* (Bouche´, 1835), is an important ectoparasite of dogs and cats throughout the world, causing annoyance to the animals and acting as a vector of infections and a cause of allergic dermatitis in dogs and cats. Although climatic variability and seasonality are known to influence the diversity and abundance of fleas, few investigations of seasonal prevalence of cat flea infestation have involved the same group of dogs being examined regularly over an extended period. The aim of this study was to determine the effects of temperature, rainfall, and relative humidity on the infestation by *C. felis felis* on 88 outdoor dogs in southeastern Brazil. The dogs, which were of mixed breed, sex, and age, were examined for ectoparasites every month during the period August 2011 to July 2012, and samples of fleas were randomly collected and identified. Meteorological data, comprising mean temperature, total rainfall, and mean relative humidity, were recorded for the calendar month prior to that in which the examinations were performed. Dogs were found to be infested only with *C. felis felis*, with a higher prevalence in the months with lowest rainfall (July, August, and September). The data obtained in this investigation can be used in control programs in order to establish an efficient strategy for environmental management and the application of insecticides, particularly during the driest months of the year based on the seasonal pattern of infestation of dogs by *C. felis felis.*

Janine K. Runfola, et al. 2015. Outbreak of Human Pneumonic Plague with Dog-to-Human and Possible Human-to-Human Transmission — Colorado, June–July 2014, Morbidity and Mortality Weekly Report (MMWR)Weekly, May 1, 2015 / 64(16) 429-434.

On July 8, 2014, the Colorado Department of Public Health and Environment (CDPHE) laboratory identified Y*ersinia pestis*, the bacterium that causes plague, in a blood specimen collected from a man (patient A) hospitalized with pneumonia. The organism had been previously misidentified as *Pseudomonas luteola* by an automated system in the hospital laboratory. An investigation led by Tri-

County Health Department (TCHD) revealed that patient A's dog had died recently with hemoptysis. Three other persons who had contact with the dog, one of whom also had contact with patient A, were ill with fever and respiratory symptoms, including two with radiographic evidence of pneumonia. Specimens from the dog and all three human contacts yielded evidence of acute *Y. pestis* infection. One of the pneumonia cases might have resulted through human-to-human transmission from patient A, which would be the first such event reported in the United States since 1924. This outbreak highlights 1) the need to consider plague in the differential diagnosis of ill domestic animals, including dogs, in areas where plague is endemic; 2) the limitations of automated diagnostic systems for identifying rare bacteria such as *Y. pestis*; and 3) the potential for milder plague illness in patients taking antimicrobial agents. Hospital laboratorians should be aware of the limitations of automated identification systems, and clinicians should suspect plague in patients with clinically compatible symptoms from whom *P. luteola* is isolated.

Fleas in Art and History

Shakespeare on Fleas, Part 1

Whether Shakespeare's works were written by Shakespeare (1564 – 1616) (London, England), or by someone else named Shakespeare, references to fleas occur there numerous times, and always negatively. Shakespeare's regard towards fleas stands in contrast to many authors, who have regarded fleas with awe, amusement, or sensuality.

A few quotes from Shakespeare mentioning fleas notably include:

King Henry V. Act 2. Scene III.

Boy: Do you not remember, a' saw a flea stick upon
Bardolph's nose, and a' said it was a black soul
burning in hell-fire?

BARDOLPH: Well, the fuel is gone that maintained that fire:
that's all the riches I got in his service.

King Henry IV. Part 1. Act 2. Scene I.

Second Carrier: I think this be the most villanous house in all
London road for fleas: I am stung like a tench.

First Carrier: Like a tench! By the mass, there is ne'er a king
christen could be better bit than I have been since
the first cock.

Twelfth Night. Act 3. Scene II.

FABIAN: We shall have a rare letter from him: but you'll not deliver't?

SIR TOBY BELCH: Never trust me, then; and by all means stir on the
youth to an answer. I think oxen and wainropes
cannot hale them together. For Andrew, if he were
opened, and you find so much blood in his liver as
will clog the foot of a flea, I'll eat the rest of
the anatomy.

FABIAN: And his opposite, the youth, bears in his visage no
great presage of cruelty.

Taming of the Shrew. Act 4. Scene III.

Tailor: She says your worship means to make
a puppet of her.

PETRUCHIO: O monstrous arrogance! Thou liest, thou thread,
thou thimble,
Thou yard, three-quarters, half-yard, quarter, nail!
Thou flea, thou nit, thou winter-cricket thou!
Braved in mine own house with a skein of thread?
Away, thou rag, thou quantity, thou remnant;
Or I shall so be-mete thee with thy yard
As thou shalt think on prating whilst thou livest!
I tell thee, I, that thou hast marr'd her gown.

Tailor: Your worship is deceived; the gown is made
Just as my master had direction:
Grumio gave order how it should be done.

Merry Wives of Windsor. Act 4. Scene II.

FORD: Master Page, as I am a man, there was one conveyed
out of my house yesterday in this basket: why may
not he be there again? In my house I am sure he is:
my intelligence is true; my jealousy is reasonable.
Pluck me out all the linen.

MISTRESS FORD: If you find a man there, he shall die a flea's death.

PAGE: Here's no man.

BIRON: Pompey is moved. More Ates, more Ates! Stir them
on! Stir them on!

DUMAIN: Hector will challenge him.

BIRON: Ay, if a' have no man's blood in's belly than will
sup a flea.

DON ADRIANO DE ARMADO: By the north pole, I do challenge thee.

COSTARD: I will not fight with a pole, like a northern man:
I'll slash; I'll do it by the sword. I bepray you,
let me borrow my arms again.

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Women catching fleas (c. 1671), by Adriaen van de Velde (1636-1672) (Amsterdam, The Netherlands), red and black chalk with brown ink drawing, 18.4x23.4cm, Cleveland Museum of Art.

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